

SMART INDIA HACKATHON 2026  
CAMPUS2INDUSTRY

**Problem Statement ID** – SIH26044

**Problem Statement Title** – Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement

**Theme** – Smart Automation

**PS Category** – Software

**Team ID** – [Enter Registered Team ID]

**Team Name** – **Campus2Industry** (Team & App Name Unified)

- CAMPUS2INDUSTRY  
Autonomous Career-Readiness & Verified Talent Pipeline

- **4-Year Progressive Journey:** Day 1 to Placement Day milestone tracking
  - **Real-Time ATS Resume Engine:** Gemini AI bullet refinement & scoring
  - **Dynamic Skill-Gap Radar:** Syllabus-to-Industry market alignment
  - **Live Hiring & Cheat Sheets:** Direct access to vetted roles & study kits

CORE PRINCIPLE: "Claim ≠ Verified Skill"

IDEA: CAMPUS2INDUSTRY

Proposed Solution

- **Unified 4-Year Career Readiness:** An AI-powered ecosystem bridging academia, verified skills, industry-grade projects, internships, assessments, and final placements.
- **Progressive Student Pipeline:** Structured path: **ORIENT (Y1) → DISCOVER (Y2) → BUILD (Y3) → PRACTICE (Y3.5) → DEPLOY (Y4).**

How It Addresses the Problem

- **Common Skill Language:** Unifies academic syllabi and live industry hiring requirements into a single dynamic taxonomy.
- **Objective Competence Diagnostics:** Instantly pinpoints what a student knows, what exact skills are missing, and gives step-by-step remediation.
- **Eligibility-Driven Matching:** Connects students to verified internships and job drives matching their actual proof-of-work.

INNOVATION & UNIQUENESS

- **Evidence-Based Skill Profile (Claim ≠ Verified Skill):**  
Replaces unsubstantiated resume claims with verifiable code proofs (GitHub repo checks, proctored quizzes, and real application builds).
- **Explainable Matchmaking Engine:**  
Transparently shows matched skills, missing delta competencies, role suitability index, and direct bridge courses.
- **Next Best Action (NBA) Recommendation Engine:**  
Autonomous guidance: *"Complete Docker containerization module to reach 90% Zoho developer cutoff".*
- **Dual-Mode AI Career Coach:**  
Integrated Google Gemini AI for mock technical interview drills and ATS resume refinement with deterministic offline fallbacks.

TECHNICAL ARCHITECTURE & COMPLETE STACK

- **Frontend Presentation:** React 19, Vite 8, Vanilla CSS Glassmorphic Design System, Lucide Icons, Recharts, Responsive Grid Architecture.
- **Backend & APIs:** Python + FastAPI, Asynchronous Concurrency, RESTful Microservices, Pydantic Schema Validation, JWT Security.
- **Database & Persistence:** PostgreSQL Relational Schema (Users, Roles, Taxonomies, Assessment Logs), Browser LocalStorage Client Vault.
- **AI & Intelligence Engine:** Google Gemini Generative AI SDK (@google/generative-ai), Structured JSON Mode, Dynamic ATS Evaluator.

IMPLEMENTATION FLOW PIPELINE

**Student Profile Ingestion → Target Career Goal → Skill Assessment → Verified Evidence → Dynamic Skill-Gap Delta → Curated Remediation Roadmap → Projects/Internships → AI Mock Interview → Placement Readiness → Digital Career Passport**

 OFFICIAL PROJECT CODING & SOURCE CODE REPOSITORY (GITHUB)

<https://github.com/faizz-hub/campustoindustry>

Includes full React 19 codebase, Gemini AI integration services, modular college notes dataset, placement cheat sheets, and deployment build files.

# FEASIBILITY AND VIABILITY

## FEASIBILITY

- **Software-Only Solution:** Zero dependency on proprietary hardware; works on any mobile or desktop browser.
- **Modular Phased Scalability:** Independent decoupling of client SPA, AI microservices, and database layers.
- **Resilient Dual-Mode Engine:** Operates dynamically via Google Gemini API and graceful deterministic fallbacks.

## POTENTIAL CHALLENGES / RISKS

- **Taxonomy Drift:** Industry frameworks evolve rapidly; static benchmarks risk obsolescence.
- **AI Response Inconsistency:** Generative models may yield variable feedback across answers.
- **Student Privacy:** Academic data and placement records require strict compliance and protection.

## RISK MITIGATION

- **Versioned Skill Taxonomies:** Admin-reviewed, periodic automated data curation.
- **Strict AI Schema Grounding:** Structured JSON output mode prevents hallucinations.
- **RBAC & Data Tokenization:** Strict role-based isolation (Student vs TPO vs Recruiter).

### ASSOCIATED APPLICATION & WEBSITE DEPLOYMENT LINKS

- Live Web Application (Production): <https://github.com/faizz-hub/campustoindustry#live-demo>
- Active Prototype Local Host: <http://localhost:5173>

STUDENTS

- Clear 4-year career direction from Day 1.
- Measurable skill growth with verified proofs.
- Real-time ATS resume builder & ATS score.
- AI mock interview prep for top tech roles.

COLLEGES / FACULTY

- Department-level placement readiness heatmap.
- Early intervention for academically weak students.
- Curriculum feedback loop from rejection analytics.
- Consolidated internship & project tracking.

INDUSTRY / RECRUITERS

- Pre-verified candidate skill profiles.
- Targeted internships & live project hiring.
- 3x faster shortlisting velocity.
- Zero logistics overhead in campus drives.

**KEY BENEFITS:** One continuous 4-year ecosystem • Personalized Next Best Action • Explainable matching • Evidence-backed Digital Career Passport  
**SUCCESS METRICS:** 40% skill-gap closure • 60% boost in ATS resume clearance • 3x internship match accuracy

 **OFFICIAL PRODUCT DEMONSTRATION VIDEO LINK**  
• Product Walkthrough Video (GitHub Hosted): <https://github.com/faizz-hub/campustoindustry#product-demo-video> (Includes 4-Year Journey, ATS Resume Creator, Notes Library, and AI Mock Interview)

## OFFICIAL / PROBLEM CONTEXT

- Smart India Hackathon 2026 Official Idea Submission Guidelines & Presentation Template.
- SIH Problem Statement SIH26044: Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement.
- National Education Policy (NEP 2020) framework on continuous skill mapping & Academic Bank of Credits (ABC).

## TECHNICAL REFERENCES

- React 19 & Vite 8 Documentation — Single-page application architecture and optimized HMR bundles.
- FastAPI Documentation — Asynchronous REST APIs, concurrency benchmarks, and Pydantic validation.
- PostgreSQL 16 Documentation — Relational data schema, index optimization, and RBAC security models.
- Google Generative AI Documentation — Gemini 2.5/3.6 Flash capabilities and JSON Mode prompting.

## RESEARCH AREAS

- **Competency-Based Skill Mapping:** Research on closing the graduate employability gap through progressive learning roadmaps.
- **Explainable AI in Talent Acquisition:** Transparent algorithm design replacing black-box screening in HR Tech.
- **Industry Skill Taxonomies:** O\*NET OnLine and European ESCO hierarchical skill classification frameworks.

## PROTOTYPE NOTE

- The working prototype is fully operational with reactive UI modules, live Gemini AI career services, and persistent client state.
- Seeded job opportunities and company cheat sheets are realistic simulations modeled after Tier-1/Tier-2 campus recruitment patterns in India.
- Corporate partner hiring feeds and institutional LMS integrations will be onboarded in Phase 2 deployment.